

Experiment no 3

Aim: Details and Working on obstacle avoiding car

Theory:

Obstacle Avoiding Car: Overview

An **Obstacle Avoiding Car** is an autonomous robotic vehicle that uses sensors to detect objects in its path and navigate around them. The car is designed to move independently, making decisions based on real-time sensor input without human intervention. The system relies on an **Arduino Uno** for control, an **ultrasonic sensor** for detecting obstacles, and a **motor driver** to control the movement of the wheels.

Key Components

1. Arduino Uno:

- Acts as the central controller, processing data from the ultrasonic sensor and controlling the motors.
- The Arduino processes sensor readings and makes decisions about the car's movement, such as whether to move forward, stop, or turn.

2. DC Motors (300rpm):

- These motors are used to power the wheels of the car. The 300 RPM speed is suitable for small robotic cars and allows for controlled movement.
- The motors enable forward, backward, and turning motions of the car, depending on the signals sent by the motor driver.

3. Ultrasonic Sensor (HC-SR04):

- The ultrasonic sensor detects objects by emitting sound waves and measuring the time it takes for them to return after hitting an obstacle.
- The distance to the object is calculated, and if it falls below a certain threshold (e.g., 20 cm), the car takes action to avoid the obstacle.

4. L293D Motor Driver:

- The L293D motor driver controls the direction and speed of the DC motors. The Arduino sends control signals to the L293D, which in turn adjusts the power going to the motors.

- This motor driver enables the car to move forward, backward, and turn by controlling the flow of current to each motor.

5. **Wheels:**

- The wheels are attached to the motors, allowing the car to move in various directions. The size and type of wheels affect the car's traction and speed.

How the Obstacle Avoiding Car Works

1. **Sensor Input:**

- The **ultrasonic sensor** continuously emits sound waves from its **trigger pin** and listens for their return on the **echo pin**.
- The **Arduino** measures the time it takes for the sound waves to bounce back and calculates the distance to the nearest object in front of the car.

2. **Decision Making:**

- Based on the calculated distance, the **Arduino** makes decisions:
 - **If the distance is greater than the threshold (safe distance)**, the car continues moving forward.
 - **If the distance is smaller than the threshold (obstacle detected)**, the Arduino triggers an action to avoid the obstacle (such as stopping or turning).

3. **Motor Control:**

- The **L293D motor driver** receives commands from the **Arduino** to control the motors' direction and speed.
 - **Move Forward:** The motors spin in a direction that moves the car forward.
 - **Turn Left/Right:** The Arduino controls one motor at a time to rotate the car in the desired direction.
 - **Stop:** The car stops by cutting power to both motors.

4. **Continuous Loop:**

- The car constantly checks the distance to obstacles and adjusts its movement in real-time.
- This creates a continuous feedback loop where the car always reacts to its environment, ensuring it avoids collisions.

Simple Operational Flow

1. Start:

- The system initializes. The Arduino and sensors are powered on, and the motors are ready for control.

2. Measure Distance:

- The **ultrasonic sensor** sends out a pulse and measures the time taken for the pulse to return. The Arduino calculates the distance to any object in front.

3. Obstacle Detection:

- **If the distance is below the threshold (e.g., 20 cm)**, an obstacle is detected:
 - The car **stops** or **turns** to avoid the obstacle.
- **If no obstacle is detected**, the car **continues moving forward**.

4. Repeat:

- The car continuously repeats this process, measuring the distance to obstacles and adjusting its movement as needed.

Example: Arduino Code for Basic Movement

```
#define TRIG_PIN 9
#define ECHO_PIN 10
#define MOTOR1_PIN1 5
#define MOTOR1_PIN2 6
#define MOTOR2_PIN1 7
#define MOTOR2_PIN2 8

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(MOTOR1_PIN1, OUTPUT);
  pinMode(MOTOR1_PIN2, OUTPUT);
  pinMode(MOTOR2_PIN1, OUTPUT);
  pinMode(MOTOR2_PIN2, OUTPUT);
  Serial.begin(9600);
}

void loop() {
```

```
long duration, distance;
```

```
// Send pulse from trigger pin
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);
```

```
// Measure time for the pulse to return
duration = pulseIn(ECHO_PIN, HIGH);
```

```
// Calculate distance
distance = duration * 0.0344 / 2;
```

```
// Print distance to Serial Monitor
Serial.print("Distance: ");
Serial.println(distance);
```

```
if (distance < 20) { // Obstacle detected
    stopMotors();    // Stop the car
    delay(500);
    turnLeft();      // Turn the car left
    delay(500);
} else {
    moveForward();   // Move forward
}
}
```

```
void moveForward() {
    digitalWrite(MOTOR1_PIN1, HIGH);
    digitalWrite(MOTOR1_PIN2, LOW);
    digitalWrite(MOTOR2_PIN1, HIGH);
    digitalWrite(MOTOR2_PIN2, LOW);
}
```

```
void stopMotors() {
    digitalWrite(MOTOR1_PIN1, LOW);
    digitalWrite(MOTOR1_PIN2, LOW);
    digitalWrite(MOTOR2_PIN1, LOW);
}
```

```
digitalWrite(MOTOR2_PIN2, LOW);  
}
```

```
void turnLeft() {  
  digitalWrite(MOTOR1_PIN1, LOW);  
  digitalWrite(MOTOR1_PIN2, LOW);  
  digitalWrite(MOTOR2_PIN1, HIGH);  
  digitalWrite(MOTOR2_PIN2, LOW);  
}
```

Conclusion:

The **Obstacle Avoiding Car** is a simple yet effective autonomous system. Using an **Arduino Uno**, **ultrasonic sensor**, and a **motor driver**, it can detect obstacles and navigate around them by controlling the motors. The system continuously monitors its surroundings and makes real-time decisions to avoid collisions. This project is an excellent introduction to robotics, offering a solid foundation for building more complex autonomous systems in the future.